



Savitribai Phule Pune University

T.Y.B.Sc.(Cyber and Digital Science)

SEMESTER-V

CDS-354

Lab Course on (CDS-351)Digital Forensics-1

Lab-Book

Name: _____

College Name: _____

Roll No: _____ Division: _____

Academic Year: _____

CERTIFICATE

This is to certify that Mr./Ms._____

Seat Number_____ of T.Y.BSC (Cyber and Digital Science) Sem-V
has successfully completed Laboratory course (CDS-351 Digital Forensics 01) in
the year_____.

He/She has scored_ mark out of 10 (For Lab Book).

Subject Teacher

H.O.D./Coordinator

Internal Examiner

External Examiner

Introduction

1. About the Workbook:

This work book is intended to be used by T.Y.B.S.c(Cyber and Digital Science) Semester V students for Cyber Laws Assignments. This workbook is designed by considering all the practical concepts / topics mentioned in syllabus.

2. The objectives of this Workbook are:

- 1)Defining the scope of the course.
- 2)To bring the uniformity in the practical conduction and implementation in all colleges affiliated to SPPU.
- 3)To have continuous assessment of the course and students.
- 4)Providing ready reference for the students during practical implementation.
- 5)Provide more options to students so that they can have good practice before facing the examination.
- 6)Catering to the demand of slow and fast learners and accordingly providing the practice assignments to them.

3. How to use this Workbook:

The workbook contains eight assignments It is mandatory for students to complete all the Assignments in given slot.

4. Instructions to the students:

Please read the following instructions carefully and follow them.

- 1) Students are expected to carry this workbook every time they come to the lab for practical.
- 2) Students should prepare for the assignment by reading the relevant material which is Mentioned in ready reference.
- 3) Instructor will specify which problems to solve in the lab during the allotted slot and Student should complete them and get verified by the instructor. However, student should Spend additional hours in Lab and at home to cover all workbook assignments if needed.

- Students will be assessed for each assignment on a scale from 0 to 5

Not done	0
Incomplete	1
Late Complete	2
Needs improvement	3
Complete	4
Well Done	5

5. Instruction to the Instructors:

Make sure that students should follow above instructions.

- Explain the assignment and related concepts using white board if required or by demonstrating the software.
- Give specific input to fill the blanks inquiries which can vary from student to student.
- Evaluate each assignment carried out by a student on a scale of 5 as specified above by ticking appropriate box.
- The value should also been don assignment completion page of the respective Lab course.

6. Instructions to the Lab administrator:

You have to ensure appropriate hardware and software is made available to each student.

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Signature of Instructor:

Assignment Completion Sheet

Assignment No	Assignment Name	Marks (out of 5)	Teacher's Sign
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3	Forensics Case Study using Encase Investigator and Autopsy		
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7	Analyzing Packets using Wire shark		
8	Using Sysinternals Tools for Network Tracking and Process Monitoring		
Total(Out of)			
Total(Out of)			

Signature of Instructor:

Assignment1: Creating a Forensic Image using FTK Imager/EncaseImager

Scenario:

You are a digital forensics investigator called to examine a computer in connection with a Cybercrime Investigation. The computer, used by the suspect, is believed to contain crucial evidence relate to the illegal activities under investigation. Your task is to create a forensic image of the computer's hard drive using FTK Imager, ensuring that the image is an accurate, unaltered replica of the original data.

Objectives:

- 1.Preserve Data Integrity:** Ensure that the data on the suspect's computer is preserved with out any modifications.
- 2.Create a Forensic Image:** Generate a bit-for-bit copy of the hard drive.
- 3.Verify Data Integrity:** Confirm that the forensic image is an exact replica of the original drive.
- 4.Document the Process:** Maintain comprehensive documentation to support the chain of custody and Legal admissibility.

Step-by-Step Procedure

1. Preparation:

Gather Tools and Equipment:

- FTK Imager (latest version).
- Write blocker (hardware or software) to prevent any modifications to the original drive.
- External storage device with sufficient capacity to store the forensic image.
- Documentation materials (notebook, camera, etc.).

Initial Assessment:

- Examine the suspect's computer to identify the type and condition of the hard drive.
- Ensure you have proper authorization and legal permissions to conduct the imaging process.

2.Document the Scene:

Photograph and Note:

- Take photographs of the computer setup, including the hard drive and connections.
- Document the make, model, and serial number of the computer and hard drive.

Record Information:

- Note the date and time of the imaging process.
- Record the names of individuals involved in the process and their roles.

3. Create the Forensic Image Using FTK Imager:**Step 1: Launch FTK Imager:**

Open FTK Imager on your workstation.

Step 2: Acquire the Source Drive:

Connect the suspect's hard drive to the workstation using a write blocker to prevent any modifications.

In FTK Imager, select File > Create Disk Image.

Step 3: Select the Source:

Choose the source drive (the suspect's hard drive) from the list of available devices.

Step 4: Choose Image Destination:

Select the destination for the forensic image (external storage device). Specify the file format (e.g.,

E01, which is a common forensic image format) and set other options as needed.

Step 5: Configure Image Options:

Set options for the imaging process, such as including all partitions and verifying the image. Choose to generate hash values (e.g., MD5, SHA-1) for the forensic image to verify its integrity later.

Step 6: Start the Imaging Process:

Click Start to begin creating the forensic image. Monitor the progress and ensure no interruptions occur during the process.

Step 7: Verify the Image:

Once the imaging process is complete, use FTK Imager to verify the integrity of the image by checking the hash values against the original drive's hash values.

4. Document and Finalize:

Hash Values:

Record the hash values generated for both the original drive and the forensic image. Ensure they match.

Chain of Custody: Update the chain of custody documentation with details of the imaging process, including who handled the drive, the imaging date and time, and the location where the image is stored.

Secure Storage:

Safely store the forensic image and original drive in a secure location, ensuring that access is restricted to authorized personnel only.

5. Reporting:

Create a Forensic Report:

Prepare a detailed report documenting the imaging process, including:

- The purpose of the imaging.
- Tools and methods used.
- Chain of custody records.
- Hash values and verification results.
- Any observations or issues encountered during the process.

Goals:

Slot 1: Creating Forensic Image

1. Launch FTK Imager/Encase Imager on your computer.
2. Select the appropriate source drive (the drive you want to create a forensic image of) and destination location (where you want to save the image).
3. Explain the importance of creating a forensic image.
4. Begin the process of creating a forensic image.
5. Document each step of the image creation process.
6. Verify that the forensic image has been successfully created.

Slot 2: Check Integrity of Data and Analyze Forensic Image

1. Discuss the significance of checking the integrity of data in forensic analysis.
2. Explain the methods used to verify the integrity of the forensic image.

3. Perform data integrity checks on the created forensic image.
4. Document the results of the integrity checks.
5. Launch FTK Imager/Encase Imager and load the created forensic image.
6. Analyze the forensic image for evidence using the software's tools.
7. Document any findings or evidence discovered during the analysis.

Outcome:

By following this procedure, you will have successfully created a forensic image of the suspect's hard drive, ensuring that the data is preserved accurately and can be analyzed without affecting the original evidence.

This process supports the integrity of the investigation and prepares the evidence for further forensic analysis or legal proceedings.

Conclusion: Summarize the importance of creating a forensic image, ensuring data integrity, and analyzing the image in computer forensics. Emphasize the role of FTK Imager/Encase Imager in this process and how it helps investigate or sin forensic examinations.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment2:Data Acquisition using USB Write Blocker+FTKImager

Objective:

To gain practical experience in performing data acquisition using a USB write blocker and FTK Imager for forensic purposes

Theory :Goals of Data Acquisition:

- **Preservation of Integrity:** Ensure that the data collected is an exact replica of the original, with no alterations or damage.
- **Completeness:** Capture all relevant data, including files, metadata, and hidden or deleted information.
- **Admissibility:** Ensure that the acquired data is admissible in legal proceedings by following proper protocols.

Types of Data Acquisition:

1. Logical Acquisition:

- **Description:** Captures files and folders as they appear in the file system. It does not capture unallocated space, slack space, or deleted files.
- **Use Case:** Suitable for cases where the focus is on specific files or folders rather than a full image of the storage device.
- **Tools:** FTK Imager, EnCase, X1 Social Discovery.

2. Physical Acquisition:

- **Description:** Captures a bit-by-bit copy of the entire storage device, including all sectors (allocated and unallocated space, slack space, and deleted files).
- **Use Case:** Essential when a comprehensive image of the entire device is required, such as in criminal investigations or detailed data analysis.
- **Tools:** FTK Imager, EnCase, dd (Linux command).

3. File System Acquisition:

- **Description:** Extracts data based on the file system structure, capturing files, directories, and metadata.
- **Use Case:** Useful when detailed analysis of the file system structure is required, often in combination with other methods.

4. Volume Acquisition:

- **Description:** Captures specific volumes or partitions on a storage device rather than the entire device.
- **Use Case:** Used when only certain partitions or volumes are of interest.

3. Data Acquisition Process:

1. Preparation: Prepare the Workstation: Ensure the forensic workstation or evidence collection system is properly set up, with forensic tools and software installed.

- **Write Protection:** Use write blockers to prevent any modifications to the original storage media during the acquisition process.

2. Collection:

- **Connect Storage Media:** Attach the storage device (e.g., hard drive, USB flash drive) to the forensic workstation via the write blocker.
- **Select Acquisition Tool:** Open the forensic imaging software (e.g., FTK Imager, EnCase).
- **Choose Acquisition Method:** Decide whether logical, physical, file system, or volume acquisition is required.
- **Start Acquisition:** Initiate the data acquisition process. The tool will create an image file or extract data based on the selected method.

3. Verification:

- **Hash Values:** Generate hash values (e.g., MD5, SHA-1) for both the original data and the acquired image. This helps verify the integrity of the data.
- **Compare Hashes:** After acquisition, compare the hash values to ensure that the image is an exact replica of the original.

4. Documentation:

- **Chain of Custody:** Document every step of the acquisition process, including details about the storage media, tools used, and the personnel involved.
- **Report:** Prepare a detailed report outlining the acquisition process, including the methods used, hash values, and any issues encountered.

4. Tools and Technologies:

- **FTK Imager:** A widely used forensic tool for creating disk images and verifying data integrity.
- **EnCase:** A comprehensive forensic suite that supports various types of data acquisition and analysis.
- **dd:** A command-line tool used for creating bit-by-bit copies of storage devices in Unix-like systems.
- **X1 Social Discovery:** Specializes in acquiring and analyzing social media and web data.

5. Legal and Ethical Considerations:

- **Legal Compliance:** Follow legal procedures and obtain proper authorization before acquiring data.
- **Privacy:** Ensure that the data acquisition process respects privacy laws and regulations.
- **Chain of Custody:** Maintain a clear chain of custody to ensure the evidence remains admissible in court.

Instructions:

Slot 1: Data Acquisition using USB Write Blocker

1. Connect the USB Write Blocker to your computer and the source drive (e.g., USB flash drive or external hard drive).
2. Explain the purpose of a USB Write Blocker in forensic data acquisition.
3. Ensure that the USB Write Blocker is properly connected and functioning as intended.
4. Document the steps you take to set up the USB Write Blocker for data acquisition.
5. Verify that write-blocking is active to prevent any modifications to the source drive.
6. Begin the process of data acquisition from the source drive using the USB Write Blocker.
7. Document the data acquisition process step by step.

Slot 2: Data Acquisition using FTK Imager

1. Launch FTK Imager on your computer.
2. Explain the role of FTK Imager in forensic data acquisition.
3. Choose the option to create a new image file.
4. Specify the source device (the drive connected via the USB Write Blocker) and the destination location for the acquired data.
5. Configure the acquisition settings, such as file format and compression, as necessary.
6. Initiate the data acquisition process using FTK Imager.
7. Document the progress and any information provided by the software during acquisition.
8. Verify the completeness and integrity of the acquired data.

Conclusion: Summarize the importance of using a USB Write Blocker and FTK Imager for data acquisition in computer forensics. Emphasize the need for maintaining the integrity of the evidence during the acquisition process and how these tools facilitate that.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 3 : Forensics Case Study using Encase Investigator and Autopsy

Objective: To apply forensic investigation techniques by solving a case study using two different forensic tools, Encase Investigator and Autopsy.

Theory :

1. Create a new case in EnCase Investigator and import the image file (evidence) into the case.

2. Conduct a preliminary examination of the evidence, including:

- Viewing file system structure and contents
- Identifying operating system and file system type
- Checking for any obvious signs of tampering or deletion

3. Use EnCase's built-in tools to:

- Recover deleted files and folders
- Reconstruct file system structures
- Extract metadata and timestamps

4. Analyze file contents, including:

- Document and image files
- Email and chat logs
- System and application logs

5. Use EnCase's filtering and sorting capabilities to:

- Identify suspicious files or activity
- Correlate evidence with case facts and timelines

6. Document and report findings, including:

- Creating a detailed report of evidence and analysis
- Highlighting key findings and conclusions
- Supporting findings with screenshots and exhibits

Forensics Case Study: Solve the Case study (image file) provide in lab using Autopsy.

1. Open Autopsy and create a new case.

2. Add the image file (evidence) to the case.

3. Conduct an initial analysis using Autopsy's built-in modules:

- File System Analysis: Explore the file system structure and contents.
- File Analysis: Examine file metadata, hashes, and contents.
- Keyword Search: Search for specific keywords or phrases.

4. Use Autopsy's plugins to:

- Extract and analyze email, chat, and social media data.
- Reconstruct system and application logs.
- Identify and extract malware.

5. Analyze file contents, including:

- Document and image files.
- Email and chat logs.
- System and application logs.

6. Use Autopsy's timeline analysis to:

- Correlate evidence with case facts and timelines.
- Identify suspicious activity.

7. Document and report findings, including:

- Creating a detailed report of evidence and analysis.
- Highlighting key findings and conclusions.
- Supporting findings with screenshots and exhibits.

Some specific steps in Autopsy include:

- Using the "File System" module to navigate the file system and identify suspicious files or directories.
- Using the "File Analysis" module to examine file metadata, hashes, and contents.
- Using the "Keyword Search" module to search for specific keywords or phrases.
- Using the "Email" and "Chat" modules to extract and analyze email and chat data.
- Using the "Timeline" module to correlate evidence with case facts and timelines.

Instructions:

Slot 1:Forensics Case Study with Encase Investigator

1. Launch Encase Investigator on your computer.
2. Open the provided case study image file.
3. Describe the details of the case study, including any background information or context.
4. Identify and document the objectives of the investigation.
5. Utilize Encase Investigator's tools and features to examine the case study image.
6. Document your findings, including any potential evidence, artifacts, or anomalies.
7. Analyze and interpret the data to draw conclusions about the case.

8. Prepare a report summarizing your investigation, findings, and conclusions.

Slot 2: Forensics Case Study with Autopsy

1. Launch Autopsy on your computer.
2. Open the same provided case study image file used in Slot 1.
3. Review the case study information and objectives.
4. Explain how Autopsy can be used in a forensic investigation
5. Use Autopsy's features and modules to analyze the case study image.
6. Document your findings, including any evidence or information discovered.
7. Analyze the data and draw conclusions based on your investigation.
8. Create a report summarizing your findings, analysis, and conclusions, similar to Slot 1.

Conclusion: Summarize the experience of solving a forensics case study using both Encase Investigator and Autopsy. Discuss the advantages and disadvantages of each tool and their role in digital forensics investigations.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 4 : Recovering and Inspecting Deleted Files

Objectives: To gain practical experience in recovering and inspecting deleted files from storage media.

Requirements:

Computer with appropriate recovery software installed (e.g., Recuva, PhotoRec)

Storage media with deleted files (you can simulate this for the assignment)

Basic understanding of file recovery and analysis techniques

Theory:

1. Check for Deleted Files

- When a file is deleted, the operating system removes the file's entry from the file system's directory structure.
- The file's contents remain on the disk until they are overwritten by new data.
- Deleted files can be recovered if the disk space has not been overwritten.

2. Recover the Deleted Files

- **File carving:** A technique used to recover files based on their content, rather than their file system metadata.
- **File system analysis:** Analyzing the file system's metadata to identify deleted files.
- **Disk imaging:** Creating a bit-for-bit copy of the disk to preserve the original data.

3. Analyzing and Inspecting the Recovered Files

- **Metadata analysis:** Examining the file's metadata, such as creation and modification dates, file size, and ownership.
- **Content analysis:** Examining the file's contents, such as text, images, or other data.
- **File signature analysis:** Identifying the file type based on its signature or magic number.
- **Hash analysis:** Comparing the file's hash value to known hash values to identify the file.

4. Tools Used

- Autopsy
- EnCase
- FTK

- Recuva
- PhotoRec

5. File Recovery Techniques

- File system-based recovery
- File carving
- Disk imaging
- Memory analysis

6. File Inspection Techniques

- Metadata analysis
- Content analysis
- File signature analysis
- Hash analysis

7. Best Practices

- Always work with a disk image or copy to avoid modifying the original data.
- Use multiple tools and techniques to verify results.
- Document all steps and findings.
- Consider the file's context and potential relevance to the investigation

Instructions:

Slot 1: Check for Deleted Files

1. Explain the importance of checking for deleted files in digital forensics investigations.
2. Launch the file recovery software (e.g., Recuva or PhotoRec) on your computer.
3. Select the storage media (e.g., a USB drive or simulated storage) to scan for deleted files.
4. Configure the recovery software settings, such as file types to recover and search parameters.
5. Start the scanning process to identify deleted files on the selected storage media.
6. Document the list of detected deleted files.

Slot 2: Recover the Deleted Files and Analyze/Inspect

1. Explain the process of recovering deleted files and its significance in digital forensics.
2. Choose and recover a selected deleted file from the list obtained in Slot 1.
3. Document the recovery process, including the file's original location and metadata.
4. Analyze and inspect the recovered file using appropriate software (e.g., a text editor for documents, an image viewer for images).
5. Examine the content of the recovered file for any relevant information or evidence.

6. Discuss the potential implications of the recovered file in a forensic investigation.
7. Summarize the findings and relevance of the recovered file in your analysis.

Conclusion: Summarize the importance of recovering and inspecting deleted files in digital forensics. Discuss how this process can contribute to a forensic investigation and the potential evidence that can be uncovered through file recovery and analysis.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 5 : Web Browser Forensics

Objectives: To explore web browser forensics, including understanding how web browsers work, conducting forensic activities on a browser, and analyzing cache and cookies.

Requirements:

Computer with appropriate recovery software installed (e.g., Recuva, PhotoRec)

Storage media with deleted files (you can simulate this for the assignment)

Basic understanding of file recovery and analysis techniques

Theory:

Web Browser Working

1. Web browsers store data in various formats, including:

- Browsing history (URLs, timestamps, titles)
- Cookies (text files with tracking info)
- Cache (temporary internet files, e.g., HTML, images)
- Bookmarks (saved URLs)
- Search history (search queries)
- Download history (downloaded files)
- Form data (auto fill information)
- Passwords (stored login credentials)

2. Forensics Activities on Browser

- Collecting and analyzing browser data for evidence
- Examining browser artifacts to reconstruct user activities
- Identifying potential security threats or malicious activities
- Analyzing browser extensions and plugins
- Recovering deleted browser data (e.g., using file carving)

3. Cache / Cookies Analysis

○ Cache analysis:

- Examining temporary internet files (e.g., HTML, images)
- Reconstructing web pages and user interactions
- Identifying potential evidence of malicious activities

○ Cookies analysis:

- Examining cookie files for tracking and authentication information

- Identifying potential security risks (e.g., session hijacking)
- Analyzing cookie expiration dates and modification times

4. Last Internet Activity

- Analyzing the last internet activity can help investigators determine:

- The user's last actions before a security incident
- Potential sources of malware or exploitation
- The user's online behavior and habits
- The timeline of events leading up to an incident

5. Browser Forensics Tools

- Google Chrome: Chrome Forensics, Chrome Cache Viewer
- Mozilla Firefox: Firefox Forensics, Firefox Cache Viewer
- Microsoft Edge/Internet Explorer: Internet Explorer Cache Viewer, Edge Forensics
- Safari: Safari Forensics, Safari Cache Viewer

6. Best Practices

- Use specialized tools and techniques for browser forensics
- Document all steps and findings
- Consider the context and potential relevance of browser data to the investigation
- Analyze multiple browser artifacts to reconstruct user activities

Instructions:

Slot 1: Web Browser Working

1. Explain the basics of how web browsers work, including their purpose and functionality.
2. Launch your web browser of choice on your computer.
3. Navigate to a few websites of your choice to demonstrate typical browser usage.
4. Document the steps you take to access and interact with websites.
5. Describe the importance of understanding web browser behavior in digital forensics.

Slot 2: Forensics Activities on Browser, Cache/Cookies Analysis

1. Discuss the concept of web browser forensics and its role in digital investigations.
2. Explore and demonstrate forensic activities on your web browser, including:
3. Checking browsing history
4. Examining bookmarks and saved passwords
5. Investigating downloaded files
6. Identifying stored form data (e.g., auto fill)

7. Explain the importance of each activity in a digital forensics context.
8. Analyze the cache and cookies of your web browser:
9. Describe what browser cache and cookies are and their functions.
10. Use browser settings or browser forensics tools to view and analyze cache and cookies.
11. Discuss how cache and cookies analysis can provide insights into a user's online activities.
12. Document any findings or relevant information discovered during the forensic activities.
13. Summarize the importance of web browser forensics in digital investigations

Conclusion: Summarize the significance of web browser forensics in digital investigations, emphasizing the role of cache and cookies analysis in uncovering user activities. Discuss how understanding web browser behavior can aid investigators in piecing together digital evidence.

Assignment Evaluation

0: Not Done []

3: Need Improvement []

1: Incomplete []

4: Complete []

2: Late Complete []

5: Well Done []

Signature of Instructor

Assignment 6 : Capturing and Analyzing Network Packets using Wireshark (Fundamentals)

Objectives: To gain a foundational understanding of capturing and analyzing network packets using Wireshark

Requirements:

Computer with Wireshark software installed

Access to a live network or simulation environment

Basic knowledge of networking concepts

Theory:

1. Identification of the Live Network

- **Network Interface Cards (NICs):** Wireshark interacts with the network through NICs. Each NIC corresponds to a specific network interface, like Ethernet, Wi-Fi, or virtual interfaces.
- **Interfaces List:** Wireshark provides a list of available interfaces on the system. This list includes physical interfaces (e.g., Ethernet adapters), virtual interfaces (e.g., VPNs), and loopback interfaces.
- **Selection Criteria:** To capture relevant data, you must select the interface that handles the traffic of interest. For instance, if you want to capture Wi-Fi traffic, select the wireless interface. For wired traffic, select the Ethernet interface.
- **Interface Status:** Check if the interface is up and running. Interfaces that are down or disabled won't capture any traffic.

2. Capture Packets

- **Packet Capture (PCAP):** Wire shark uses the PCAP (Packet Capture) library to capture network packets. The library interacts with the network stack of the operating system to intercept packets in real-time.
- **Capture Filters:** Before starting a capture, you can apply filters to limit the traffic collected. Capture filters are set using the Berkeley Packet Filter (BPF) syntax and allow you to capture only specific packets (e.g., only HTTP traffic or packets from a specific IP address).
- **Starting the Capture:** Once the desired interface is selected, clicking the “Start Capturing Packets” button will begin the process of collecting packets from the network. The packets will be displayed in real-time in Wire shark's user interface.

- **Stopping the Capture:** Capturing can be stopped at any time by clicking the “Stop” button. This will end the packet capture session and allow you to analyze the collected data.

3. Analyze the Captured Packets

A. Packet Details: Each packet captured consists of several layers, including:

- **Ethernet Layer:** Contains MAC addresses and other Ethernet-specific information.
- **IP Layer:** Contains IP addresses and related information (e.g., protocol type).
- **Transport Layer:** Includes TCP/UDP headers with port numbers and flags.
- **Application Layer:** Contains the payload or data from applications (e.g., HTTP requests/responses).

B. Packet Inspection: Wireshark allows detailed inspection of each packet, showing headers and payloads. You can drill down into each layer to analyze its contents.

C. Filters: Wireshark uses display filters to help narrow down the packets of interest after capture. Display filters use a syntax similar to capture filters but are applied to already captured data (e.g., http to show only HTTP traffic or ip.addr == 192.168.1.1 to show packets to/from a specific IP).

D. Statistics and Analysis: Wireshark provides tools to analyze traffic patterns, such as:

- **Protocol Hierarchy:** Shows a breakdown of protocols used in the capture.
- **Conversations:** Lists all network conversations, such as those between two IP addresses.
- **Endpoints:** Displays statistics on network endpoints, including IP addresses and MAC addresses.
- **IO Graphs:** Provides graphical representations of traffic over time to visualize network behavior.

Instructions:

Slot 1: Identifying the Live Network

1. Explain the importance of identifying the live network in network packet analysis.
2. Launch Wireshark on your computer.
3. Describe the interface and various components of the Wireshark application.
4. Use Wireshark to detect and identify the live network interfaces available on your computer.
5. Document the information about the identified live network interfaces.

Slot 2: Capturing Packets and Analyzing Captured Packets

1. Discuss the process of capturing network packets and its relevance in network forensics.
2. Select one of the identified live network interfaces to capture packets.

3. Configure Wireshark capture settings, such as capture filters and display filters, as needed.
4. Start the packet capture process and let it run for a brief period.
5. Document the number of packets captured during the capture session.
6. Stop the packet capture and save the captured data as a Wireshark capture file.
7. Open the captured file in Wireshark for analysis.
8. Analyze the captured packets by examining their contents, source/destination addresses, protocols, and any anomalies.
9. Document any interesting findings, potential issues, or suspicious activities discovered in the captured packets.
10. Summarize the importance of capturing and analyzing network packets in network forensics and troubleshooting.

Conclusion: Summarize the significance of capturing and analyzing network packets using Wireshark, highlighting its role in network forensics and network troubleshooting. Emphasize the value of understanding network traffic for security and investigative purposes.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 7 : Analyzing Packets using Wireshark

Objectives: To practice packet analysis using Wireshark to answer specific questions based on provided packet data.

Requirements: Packet capture file (provided in the lab) Wireshark software installed on your computer Basic knowledge of using Wireshark for packet analysis

1. What Web Server Software is Used by www.snopes.com?

Objective: Identify the web server software used by a specific website.

Theory and Steps:

- **HTTP Headers:** Web servers typically provide information about their software in the HTTP headers of responses. Look for the Server header in HTTP responses.
- **Steps in Wireshark:**
 1. **Filter HTTP Traffic:** Use a display filter to focus on HTTP traffic. Apply the filter http.
 2. **Find Relevant Packets:** Look through the captured HTTP packets to find a response from www.snopes.com. You might filter by `http.host == "www.snopes.com"`.
 3. **Inspect HTTP Headers:** Open the packet details and expand the Hypertext Transfer Protocol section to examine the Server field. This field usually contains the name and version of the web server software.

2. About What Cell Phone Problem is the Client Concerned?

Objective: Determine the specific cell phone issue discussed in the captured network traffic.

Theory and Steps:

- **Application Layer Traffic:** The problem might be discussed through an application protocol like HTTP, which can be found in the payload of the packets.
- **Steps in Wireshark:**
 1. **Filter HTTP Traffic:** Apply the filter http to narrow down to HTTP traffic.
 2. **Find Relevant Packets:** Look for HTTP packets that might contain textual data related to the problem. You might need to examine HTTP POST requests or responses for form submissions or content.

3. **Inspect Payload:** Look at the payload of the relevant HTTP requests or responses. The problem might be described in a form submission, email body, or message content. Expand the packet details to view the HTTP data.

3. According to Zillow, What Instrument Will Ryan Learn to Play?

Objective: Find out which instrument Ryan will learn to play according to information from Zillow.

Theory and Steps:

- **Application Layer Traffic:** Similar to the cell phone problem, the information might be conveyed through application-specific protocols like HTTP.
- **Steps in Wireshark:**
 1. **Filter HTTP Traffic:** Use http as your display filter to focus on HTTP packets.
 2. **Find Relevant Packets:** Look for HTTP responses from zillow.com. You might filter by `http.host == "www.zillow.com"`.
 3. **Inspect Payload:** Look at the content of HTTP responses for information regarding the instrument. The details might be in a web page or data exchange.

4. How Many Web Servers Are Running Apache?

Objective: Determine the number of web servers identified as running Apache.

Theory and Steps:

- **Server Identification:** The Server HTTP header often includes details about the web server software, including Apache.
- **Steps in Wireshark:**
 1. **Filter HTTP Traffic:** Apply the filter http.
 2. **Find Relevant Packets:** Look for HTTP response packets across various hosts. Use a filter like `http.server` to find responses with the Server header.
 3. **Count Apache Servers:** Examine the Server header in the HTTP responses. Count how many different responses indicate the presence of Apache.
- **Summary of the Approach**
 1. **Filter the traffic** based on the protocol or host related to the question (HTTP for web-related queries).
 2. **Examine the packet details** carefully, especially HTTP headers and payloads, to find the required information.
 3. **Count and summarize** the results based on what you find in the headers or content.

Instructions:

Slot 1: Analyzing Packets and Answering Questions

1. Launch Wireshark on your computer.
2. Open the provided packet capture file.
3. Review the questions that need to be answered based on the packet data.
4. Begin by finding the web server software used by www.snopes.com.
5. Use Wireshark's filter and search features to identify relevant packets.
6. Document the web server software used.
7. Continue by identifying the cell phone problem the client is concerned about.
8. Analyze the packet data to find relevant information about the cell phone problem.
9. Document the client's concern.
10. Proceed to find out what instrument Ryan will learn to play according to Zillow.
11. Examine the packet data for clues regarding Ryan's instrument.
12. Document the instrument mentioned.
13. Lastly, determine how many web servers are running Apache.
14. Use Wireshark to count the instances of Apache server responses in the captured packets.
15. Document the number of Apache web servers detected.

Slot 2: Analyzing Packets and Answering Questions (Continued)

1. Review and verify your findings from Slot 1.
2. If needed, revisit the packet data to ensure accuracy in your answers.
3. Document any additional details or insights related to the questions.
4. Summarize your findings and answers to each of the questions.
5. Reflect on the importance of packet analysis in gaining insights into network communication.

Conclusion: Summarize the experience of analyzing packets using Wireshark to answer specific questions. Emphasize the role of packet analysis in network troubleshooting and forensic investigations.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor

Assignment 8 : Using Sysinternals Tools for Network Tracking and Process Monitoring

Objectives: To explore and utilize Sysinternals tools for network tracking and process monitoring on a Windows-based computer.

Requirements:

- A Windows-based computer with Sysinternals Suite installed (downloadable from Microsoft)
- Basic knowledge of computer processes and network tracking

Theory:

1. Check Sysinternals Tools

- **Sysinternals Suite:** This is a collection of system utilities that provide detailed information and control over various aspects of a Windows operating system. The suite includes tools for monitoring processes, network activity, file systems, and more.
- **Popular Tools:**
 1. **Process Explorer:** Provides detailed information about running processes, including their resource usage, and allows you to manage processes.
 2. **Process Monitor:** Captures real-time file system, registry, and process/thread activity.
 3. **TCP View:** Displays a list of all TCP and UDP endpoints on your system, showing which applications are using network connections.
 4. **RAM Map:** Analyzes physical memory usage on your system, including details on memory allocation.
 5. **Sysmon:** Monitors and logs system activity, including process creation, network connections, and file creation.

2. Monitor Live Processes

Steps:

- **Process Explorer:** Use this tool to view and manage live processes.
 - **Process Tree:** Displays a hierarchical view of processes, showing parent-child relationships.
 - **Process Details:** Provides information such as CPU usage, memory usage, handles, and threads.
 - **Process Management:** Allows you to terminate or suspend processes and view detailed properties.
 - **Real-Time Monitoring:** Offers live updates on process activities and resource consumption.

3. Capture RAM

Steps:

- **RAM Map:** This tool provides detailed insights into memory usage, including:
 - **Memory Usage Breakdown:** Shows how memory is allocated and used by various system components.
 - **Physical Memory:** Provides an overview of free, active, and modified memory.
 - **File Summary:** Displays information about memory mapped files and their usage.
 - **Capture Memory Snapshot:** You can analyze snapshots of memory usage to understand allocation patterns and detect anomalies.

4. Capture TCP/UDP Packet

Steps:

- **TCP View:** This tool displays detailed information about current TCP and UDP connections.
 - **Connection Details:** Shows local and remote addresses, connection states, and the process associated with each connection.
 - **Real-Time Updates:** Provides a live view of active connections and their status.
 - **Filtering and Analysis:** Allows you to filter connections and analyze specific endpoints.

5. Monitor Hard Disk

Steps:

- **Process Monitor:** Provides insights into file system activity, which includes hard disk read/write operations.
 - **File System Activity:** Logs access to files and directories, including reads, writes, and file operations.
 - **Performance Monitoring:** You can analyze the performance impact of disk operations and identify potential bottlenecks.
- **Disk Usage Tools:** Sysinternals provides tools like **Disk Usage (du)**, which can be used to analyze disk space usage by directories and files.

6. Monitor Virtual Memory

Steps:

- **Sysinternals Tools:**
 - **RAM Map:** Provides a breakdown of virtual memory usage, including details on paged and non-paged pool memory.

- **System Resource Usage:** Monitors how virtual memory is being used, including details on paging file usage.
- **Performance Counters:** Use Windows Performance Monitor (PerfMon) alongside Sysinternals tools to track virtual memory metrics.

7. Monitor Cache Memory

Steps:

- **RAM Map:** Provides insights into cache memory usage:
 - **Cache Overview:** Displays information about cached files and system caches.
 - **Cache Breakdown:** Shows how memory is allocated to different types of caches, including file system cache and kernel cache.
 - **Efficiency Analysis:** Helps analyze how efficiently cache memory is being utilized and identify potential optimization areas.

Summary

- **Sysinternals Tools:** Offer powerful capabilities for detailed system monitoring and analysis.
- **Process Monitoring:** Use tools like Process Explorer to manage and observe live processes.
- **RAM Analysis:** Tools like RAM Map provide insights into physical memory usage.
- **Network Traffic:** TCP View helps capture and analyze network packets and connections.
- **Disk and Memory Monitoring:** Track disk activity and virtual/cache memory using tools like RAM Map and Process Monitor.

Instructions:

Slot 1: Checking Sysinternals Tools and Monitoring Live Processes

1. Explain the purpose and significance of Sysinternals tools in system monitoring.
2. Launch the Sysinternals Suite on your computer.
3. Explore the list of Sysinternals tools available within the suite.
4. Choose a specific tool (e.g., Process Explorer) for monitoring live processes.
5. Demonstrate how to use the selected tool to view and monitor running processes on your computer.
6. Document any findings or observations related to live processes.

Slot 2: Capturing RAM, TCP/UDP Packets, Hard Disk, Virtual Memory, and Cache Memory

1. Discuss the various aspects of system monitoring covered by Sysinternals tools, including RAM, network traffic (TCP/UDP packets), hard disk activity, virtual memory, and cache memory.
2. Select appropriate Sysinternals tools for capturing and monitoring each of these aspects.
3. Use Sysinternals tools to:

4. Capture and analyze RAM usage and processes.
5. Capture and inspect TCP/UDP packets in real-time.
6. Monitor hard disk activity and analyze disk usage.
7. Check virtual memory utilization.
8. Monitor cache memory.
9. Document your observations, including any unusual or noteworthy behavior in each aspect.
10. Summarize the role of Sysinternals tools in system monitoring and troubleshooting.

Conclusion: Summarize the experience of using Sysinternals tools for network tracking and process monitoring on a Windows computer. Emphasize the value of these tools in gaining insights into system behavior and performance.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Need Improvement []

4: Complete []

5: Well Done []

Signature of Instructor